

The L^q , $p = 1$ Endpoint for $1 < q \leq 2$

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Status

This packet gives a candidate partial solution to Remark 12.10 of Yaroslavtsev's paper *Local characteristics and tangency of vector-valued martingales* [1]. The source asks whether characteristic domination implies maximal L^p comparison for general local martingales. Earlier packets in this run prove the $L^q(S)$ -valued case for $1 < p, q < \infty$, the Hilbert case for all finite p , and a conditional reduction of the remaining $L^q(S)$, $p = 1$ endpoint to one square-function comparison.

The new result here proves that square-function comparison for $1 < q \leq 2$. Consequently the full $L^q(S)$ -valued characteristic-domination conclusion holds at $p = 1$ for $1 < q \leq 2$. The result remains partial relative to the source problem: $q > 2$ and arbitrary UMD spaces are not settled here.

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. The source states that the general local-martingale case is open and reduces the missing issue to an independent-increment comparison.

Now we try to prove the case of discrete martingales. By Theorem (12.11), (12.14), (2.9), and the fact that M and N are quasi-left continuous. $\square$

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n \geq 1}$ and $(\eta_n)_{n \geq 1}$ be X -valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exists a constant C (perhaps depending on p and X) such that

$$\mathbb{E} \left\| \sum_n \xi_n \right\|^p \leq C \mathbb{E} \left\| \sum_n \eta_n \right\|^p?$$

The continuation states the aggregate-domination formulation and notes that even the symmetric case was not known to the author.

By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n 's and η_n 's are symmetric. But even the symmetric case is not known for the author.

Main Result

For a finite independent family $x = (x_i)$ of $L^q(S)$ -valued random variables, define

$$\text{SF}_q(x) = \mathbb{E} \left\| \left(\sum_i |x_i|^2 \right)^{1/2} \right\|_{L^q(S)}.$$

Let

$$\mu_x = \sum_i \mathcal{L}(x_i)|_{L^q(S) \setminus \{0\}}$$

be the aggregate occurrence measure.

Theorem 1 (Endpoint square-function comparison, $1 < q \leq 2$). *Let $1 < q \leq 2$. There is an absolute constant $K \leq e/(e-1)$ such that for every measure space S and every two finite independent families $x = (x_i)$, $y = (y_j)$ of $L^q(S)$ -valued random variables,*

$$\mu_x \leq \mu_y \implies \text{SF}_q(x) \leq K \text{SF}_q(y).$$

Theorem 2 (General local L^q , $p = 1$, $1 < q \leq 2$). *Let $1 < q \leq 2$, let S be a measure space, and put $X = L^q(S)$. If N and M are X -valued local martingales and N is characteristically dominated by M in the sense of [1, Definition 12.9], then*

$$\mathbb{E} \sup_{t \geq 0} \|N_t\|_{L^q(S)} \leq C_q \mathbb{E} \sup_{t \geq 0} \|M_t\|_{L^q(S)}.$$

Idea of the Proof

The point of the endpoint $p = 1$ is that Yaroslavtsev's Banach-function-space BDG theorem compares martingale maximal functions with lattice square functions even at $p = 1$ [2]. Thus the remaining question is whether

$$\mathbb{E} \left\| \left(\sum_i |x_i|^2 \right)^{1/2} \right\|_q$$

is monotone, up to an absolute constant, under aggregate domination of the individual jump laws.

For $1 < q \leq 2$ this square-function norm has exactly the right Laplace structure. On a finite weighted ℓ_q^d sublattice, put

$$r = q/2, \quad \Phi(v) = \left(\sum_{k=1}^d a_k v_k^r \right)^{1/q}, \quad v \in \mathbb{R}_+^d.$$

Since $r \leq 1$, each v_k^r is the Laplace exponent of a stable subordinator. Since $1/q < 1$, the outer power is again a Bernstein operation. Therefore

$$\Phi(v) = \int_{\mathbb{R}_+^d \setminus \{0\}} \left(1 - e^{-\langle t, v \rangle}\right) \Pi(dt)$$

for a positive measure Π . For independent positive vectors V_i , the expectation of the kernel is

$$1 - \prod_i \mathbb{E} e^{-\langle t, V_i \rangle}.$$

Writing $u_i(t) = \mathbb{E}(1 - e^{-\langle t, V_i \rangle})$, this quantity is trapped between constants times $\min(1, \sum_i u_i(t))$. The sum $\sum_i u_i(t)$ depends only monotonically on the aggregate occurrence measure. This is the whole mechanism.

The Finite-Dimensional Laplace Lemma

Lemma 1 (Bernstein representation). *Let $1 < q \leq 2$, let $r = q/2$, let $a_1, \dots, a_d > 0$, and set*

$$\Phi(v) = \left(\sum_{k=1}^d a_k v_k^r \right)^{1/q}, \quad v \in \mathbb{R}_+^d.$$

There exists a positive measure Π on $\mathbb{R}_+^d \setminus \{0\}$ such that, for all $v \in \mathbb{R}_+^d$,

$$\Phi(v) = \int_{\mathbb{R}_+^d \setminus \{0\}} \left(1 - e^{-\langle t, v \rangle}\right) \Pi(dt).$$

Proof. Put $\alpha = 1/q \in (0, 1)$ and $H(v) = \sum_{k=1}^d a_k v_k^r$. The scalar identity

$$s^\alpha = c_\alpha \int_0^\infty (1 - e^{-\lambda s}) \lambda^{-1-\alpha} d\lambda, \quad s \geq 0,$$

holds for $0 < \alpha < 1$. Hence

$$\Phi(v) = c_\alpha \int_0^\infty \left(1 - e^{-\lambda H(v)}\right) \lambda^{-1-\alpha} d\lambda.$$

For $0 < r < 1$, $e^{-\lambda a_k v_k^r}$ is the Laplace transform of the positive r -stable subordinator at time λa_k ; for $r = 1$ it is the Laplace transform of the point mass at λa_k . Therefore, for each $\lambda > 0$, there is a probability measure ρ_λ on $\mathbb{R}_+^d$ such that

$$e^{-\lambda H(v)} = \int_{\mathbb{R}_+^d} e^{-\langle t, v \rangle} \rho_\lambda(dt).$$

Thus

$$1 - e^{-\lambda H(v)} = \int_{\mathbb{R}_+^d} \left(1 - e^{-\langle t, v \rangle}\right) \rho_\lambda(dt).$$

Integrating the measures $c_\alpha \lambda^{-1-\alpha} \rho_\lambda d\lambda$ over $\lambda > 0$ gives the required positive measure Π . The origin may be discarded because the kernel vanishes at $t = 0$. $\square$

Lemma 2 (Finite-dimensional aggregate comparison). *Let $1 < q \leq 2$. Let V_i, W_j be independent $\mathbb{R}_+^d$ -valued random vectors. Suppose that their aggregate occurrence measures satisfy*

$$\sum_i \mathbb{P}(V_i \in A) \leq \sum_j \mathbb{P}(W_j \in A), \quad A \in \mathcal{B}(\mathbb{R}_+^d), \quad 0 \notin A.$$

Then, for Φ as in Lemma 1,

$$\mathbb{E}\Phi\left(\sum_i V_i\right) \leq \frac{e}{e-1} \mathbb{E}\Phi\left(\sum_j W_j\right).$$

Proof. Using Lemma 1 and Tonelli's theorem,

$$\mathbb{E}\Phi\left(\sum_i V_i\right) = \int \left[1 - \prod_i \mathbb{E}e^{-\langle t, V_i \rangle}\right] \Pi(dt).$$

For fixed t , define

$$u_i(t) = \mathbb{E}\left(1 - e^{-\langle t, V_i \rangle}\right), \quad U_V(t) = \sum_i u_i(t),$$

and define $U_W(t)$ similarly. Since $0 \leq u_i(t) \leq 1$,

$$1 - \prod_i (1 - u_i(t)) \leq \min(1, U_V(t)).$$

Also $\prod_i (1 - u_i(t)) \leq e^{-U_V(t)}$, so

$$1 - \prod_i (1 - u_i(t)) \geq 1 - e^{-U_V(t)} \geq (1 - e^{-1}) \min(1, U_V(t)).$$

Aggregate domination gives

$$U_V(t) = \int_{\mathbb{R}_+^d \setminus \{0\}} (1 - e^{-\langle t, v \rangle}) d\nu_V(v) \leq \int_{\mathbb{R}_+^d \setminus \{0\}} (1 - e^{-\langle t, v \rangle}) d\nu_W(v) = U_W(t).$$

Consequently the integrand for the V_i 's is at most $(1 - e^{-1})^{-1}$ times the integrand for the W_j 's. Integrating with respect to Π proves the claim. $\square$

Proof of the Square-Function Theorem

Proof of Theorem 1. First assume that $L^q(S)$ is a finite weighted ℓ_q^d space with weights $a_k > 0$. For $z \in \ell_q^d$, set

$$Tz = (|z_1|^2, \dots, |z_d|^2) \in \mathbb{R}_+^d.$$

The aggregate domination $\mu_x \leq \mu_y$ pushes forward under T , so the positive vectors $V_i = Tx_i$ and $W_j = Ty_j$ satisfy the hypothesis of Lemma 2. Moreover

$$\left\| \left(\sum_i |x_i|^2 \right)^{1/2} \right\|_{\ell_q(a)} = \Phi \left(\sum_i V_i \right),$$

with $\Phi(v) = (\sum_k a_k v_k^{q/2})^{1/q}$. Lemma 2 therefore proves the finite-dimensional claim.

For general $L^q(S)$, use the standard finite-conditional-expectation approximation. Since the finite random families are Bochner measurable, their essential ranges are separable. After restricting to the sigma-finite support generated by a countable dense subset of these ranges, choose positive finite-rank conditional expectations P_n onto finite measurable partitions such that $P_n z \rightarrow z$ in $L^q(S)$ for every vector in those essential ranges.

The deterministic map $P_n : L^q(S) \rightarrow P_n L^q(S)$ preserves aggregate domination:

$$\sum_i \mathbb{P}(P_n x_i \in B) = \mu_x(P_n^{-1} B) \leq \mu_y(P_n^{-1} B) = \sum_j \mathbb{P}(P_n y_j \in B)$$

for every Borel $B \subset P_n L^q(S) \setminus \{0\}$. Hence the finite-dimensional result gives

$$\mathbb{E} \left\| \left(\sum_i |P_n x_i|^2 \right)^{1/2} \right\|_q \leq \frac{e}{e-1} \mathbb{E} \left\| \left(\sum_j |P_n y_j|^2 \right)^{1/2} \right\|_q.$$

For every deterministic finite family (z_i) ,

$$\left(\sum_i |P_n z_i|^2 \right)^{1/2} \leq P_n \left(\left(\sum_i |z_i|^2 \right)^{1/2} \right)$$

pointwise, by Jensen's inequality for the Hilbert norm. Thus the right-hand side is bounded by

$$\frac{e}{e-1} \mathbb{E} \left\| \left(\sum_j |y_j|^2 \right)^{1/2} \right\|_q.$$

On the other hand, $P_n x_i \rightarrow x_i$ in L^q almost surely along the chosen essential ranges, so the corresponding square functions converge almost surely in L^q . Fatou's lemma gives the desired inequality after letting $n \rightarrow \infty$. $\square$

Discrete Martingale Consequence

Proposition 1 (Discrete $p = 1$ comparison for $1 < q \leq 2$). *Let $1 < q \leq 2$. Let $(\xi_i)_{i=1}^n$ and $(\eta_j)_{j=1}^m$ be independent mean-zero $L^q(S)$ -valued random variables. If*

$$\sum_i \mathbb{P}(\xi_i \in B) \leq \sum_j \mathbb{P}(\eta_j \in B) \quad (B \in \mathcal{B}(L^q(S)), 0 \notin B),$$

then

$$\mathbb{E} \max_{1 \leq k \leq n} \left\| \sum_{i=1}^k \xi_i \right\|_{L^q(S)} \leq C_q \mathbb{E} \max_{1 \leq k \leq m} \left\| \sum_{j=1}^k \eta_j \right\|_{L^q(S)}.$$

Proof. The aggregate hypothesis is exactly $\mu_\xi \leq \mu_\eta$. Therefore Theorem 1 gives

$$\mathbb{E} \left\| \left(\sum_i |\xi_i|^2 \right)^{1/2} \right\|_{L^q} \leq \frac{e}{e-1} \mathbb{E} \left\| \left(\sum_j |\eta_j|^2 \right)^{1/2} \right\|_{L^q}.$$

Yaroslavtsev's Banach-function-space BDG theorem [2, Theorem 4.32 and Proposition 4.33] says that for every $L^q(S)$ -valued martingale difference sequence $d = (d_i)$,

$$\mathbb{E} \sup_N \left\| \sum_{i=1}^N d_i \right\|_{L^q} \simeq_q \mathbb{E} \left\| \left(\sum_i |d_i|^2 \right)^{1/2} \right\|_{L^q}.$$

Both independent mean-zero families are martingale difference sequences in any fixed order. Applying this equivalence to (ξ_i) and (η_j) proves the maximal comparison. $\square$

General Local Martingales

Proof of Theorem 2. The conditional endpoint packet in this run proves the following reduction: for a fixed $1 < q < \infty$, if the endpoint square-function aggregate comparison of Theorem 1 holds in $L^q(S)$, then the full $L^q(S)$ -valued general-local characteristic-domination theorem holds at $p = 1$. We recall the argument.

Yaroslavtsev's decoupled tangent construction and canonical decomposition reduce the comparison to continuous parts and purely discontinuous tangent jump martingales. Continuous parts are controlled at $p = 1$ by the source's continuous characteristic-domination estimate [1, Proposition 12.6]. For the jump part, conditional on the local characteristics and after the usual truncation of time and jump size, the decoupled tangent jumps are finite independent mean-zero families. Characteristic domination controls the total spatial jump compensator,

$$\nu^N(\mathbb{R}_+ \times B) \leq \nu^M(\mathbb{R}_+ \times B), \quad B \in \mathcal{B}(L^q(S)), \quad 0 \notin B,$$

which is exactly the aggregate occurrence-measure domination used in Proposition 1.

Applying Proposition 1 conditionally, then passing through the truncation and localization limits, gives the L^1 maximal comparison for the discontinuous tangent martingales. Combining this with the continuous part and the $p = 1$ Banach-function-space BDG/canonical-decomposition equivalence [2, Section 4.3] yields the zero-initial-value case. If the martingales have nonzero initial values, characteristic domination gives $\|N_0\|_{L^q} \leq \|M_0\|_{L^q}$; applying the zero-start estimate to $N - N_0$ and $M - M_0$ finishes the proof. $\square$

Why the Argument Stops at $q = 2$

The proof is not merely missing a cosmetic extension. Its central positive Laplace representation relies on $r = q/2 \leq 1$. For $q > 2$, the candidate

$$v \mapsto \left(v_1^{q/2} + v_2^{q/2} \right)^{1/q}$$

is no longer a Bernstein function on the positive cone. For example at $q = 4$, $\partial_{v_1}(v_1^2 + v_2^2)^{1/4}$ increases from 0 as v_1 leaves 0 with $v_2 > 0$ fixed, whereas a Bernstein-function partial derivative would be completely monotone. Thus the current proof cannot simply be pushed past $q = 2$.

Other routes tried during the push also appear to need a new idea for $q > 2$. Dualizing

$$\left\| \sum_i |x_i|^2 \right\|_{q/2}$$

introduces a random supremum over $L^{q/(q-2)}$ weights, while scalar Laplace comparison only handles each fixed weight. Stable-type representations of ℓ_r norms for $1 < r \leq 2$ introduce signed scalar sums; aggregate domination is not enough to compare signed fractional moments because cancellation can make the dominating family smaller. These failures are recorded as limitations, not as counterexamples.

Verification and Novelty Notes

The finite-dimensional proof is a direct positive-kernel calculation and works in extended expectation values. The passage to general $L^q(S)$ uses only positive finite-rank conditional expectations and Fatou's lemma.

Cheap run indexes contained earlier packets for the $p > 1$ L^q case, Hilbert all- p case, ℓ_q natural-exponent case, c_0 counterexamples, and the conditional $p = 1$ reduction. No packet proved the $L^q(S)$, $p = 1$, $1 < q \leq 2$ endpoint. Web searches for exact phrases around Remark 12.10, $p = 1$, L^q , characteristic domination, aggregate domination, and square functions did not reveal a later paper settling this endpoint.

Human Review Recommendation

Review Lemmas 1 and 2 first. If the multivariate Bernstein representation is accepted, the square-function comparison is immediate. Then check the finite-conditional-expectation approximation in Theorem 1; the martingale consequence is the previous conditional reduction with its conditional square-function lemma now proved for $1 < q \leq 2$.

References

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